

Hybrid OPD: Quick Results and Formalization

Setup

We compare GRPO, OPD with an SFT teacher, OPD with an RL teacher, and the success-gated hybrid OPD variant. All plotted hybrid runs use gate threshold $\tau = 0.0$ (`thr0p0` in the run name), so any positive trajectory reward selects the environment/GRPO branch. Plots report same-task and held-out task success over checkpoint step. When multiple seeds share the same run name except for seed, the line is the seed mean and the shaded band is a Gaussian 95% confidence interval.

Short Absolute-Metric Table

Table 1: Base, SFT-teacher, and GRPO step-60 absolute success metrics. The trained task column is highlighted.

Training Task	Model	Task 0	Task 1	Task 2	Task 3	Task 4	TRN AVG	HLD AVG
task_0	Base	0.66 $\pm$ 0.02	0.20 $\pm$ 0.02	0.89 $\pm$ 0.01	0.65 $\pm$ 0.01	0.32 $\pm$ 0.02	0.54 $\pm$ 0.01	0.40 $\pm$ 0.00
task_1	Base	0.66 $\pm$ 0.02	0.20 $\pm$ 0.02	0.89 $\pm$ 0.01	0.65 $\pm$ 0.01	0.32 $\pm$ 0.02	0.54 $\pm$ 0.01	0.40 $\pm$ 0.00
task_1	SFT teacher eval (task 1)	0.00	0.95	0.00	0.00	0.00	0.19	0.00
task_1	GRPO step 60	0.67	0.75	0.89	0.68	0.44	0.69	0.43
task_4	Base	0.66 $\pm$ 0.02	0.20 $\pm$ 0.02	0.89 $\pm$ 0.01	0.65 $\pm$ 0.01	0.32 $\pm$ 0.02	0.54 $\pm$ 0.01	0.40 $\pm$ 0.00
task_4	SFT teacher eval (task 4)	0.01	0.00	0.09	0.15	0.87	0.22	0.02
task_4	GRPO step 60	0.49	0.25	0.86	0.76	0.77	0.63	0.43

Hybrid OPD Formalization

For a trajectory i , define the environment success score

$$S_i = \sum_t r_{i,t}.$$

GRPO computes a group-relative environment advantage over group G :

$$A_i^{\text{env}} = \frac{S_i - \mu_G}{\sigma_G + \epsilon}, \quad \mu_G = \frac{1}{|G|} \sum_{j \in G} S_j.$$

OPD gives dense teacher feedback from teacher/student log-probability margins:

$$m_{i,t,k} = \log \pi_T(a_{i,t,k} \mid s_{i,t}) - \log \pi_{\theta_{\text{old}}}(a_{i,t,k} \mid s_{i,t}), \quad A_{i,t,k}^T = \text{Norm}(m_{i,t,k}).$$

The success gate uses threshold τ and teacher weight λ :

$$g_i = \mathbf{1}[S_i > \tau], \quad A_{i,t,k}^{\text{hyb}} = g_i A_i^{\text{env}} + (1 - g_i) \lambda A_{i,t,k}^T.$$

In the experiments shown here, $\tau = 0.0$. Thus $g_i = 1$ whenever the trajectory obtains any positive environment reward, and $g_i = 0$ for zero-reward failures. The actor update uses the GRPO/PPO clipped objective:

$$\rho_{i,t,k}(\theta) = \exp(\log \pi_{\theta}(a_{i,t,k} \mid s_{i,t}) - \log \pi_{\theta_{\text{old}}}(a_{i,t,k} \mid s_{i,t})),$$

$$\mathcal{L}_{\text{hyb}}(\theta) = \mathbb{E}_{i,t,k} \left[\max \left(-A_{i,t,k}^{\text{hyb}} \rho_{i,t,k}, -A_{i,t,k}^{\text{hyb}} \text{clip}(\rho_{i,t,k}, 1 - \epsilon_{\ell}, 1 + \epsilon_h) \right) \right].$$

In words: successful rollouts use environment/GRPO credit; failed rollouts use SFT-teacher dense OPD credit.

Checkpoint Results

Task 1: checkpoint success metrics

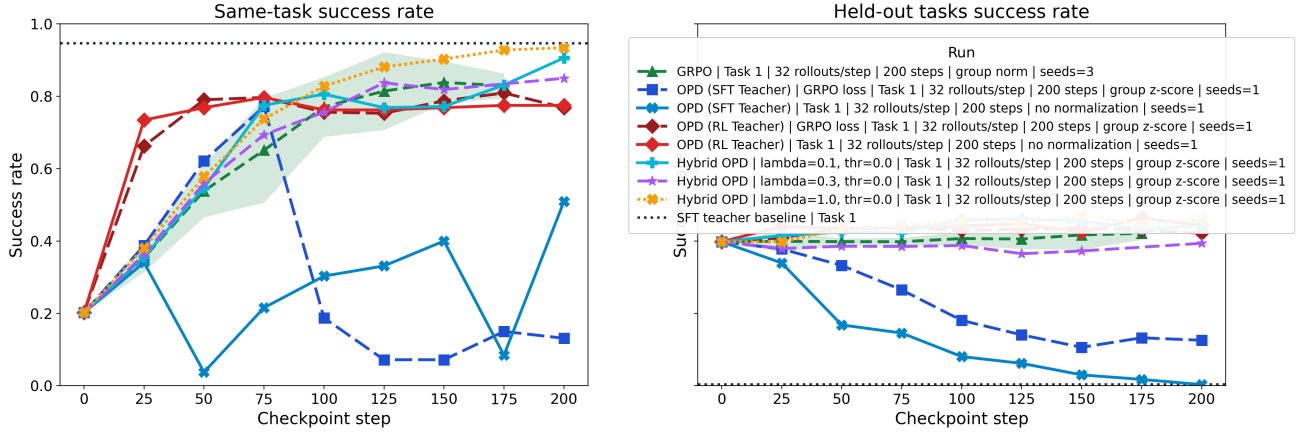


Figure 1: rps32 comparison, Task 1. Hybrid OPD uses success-gated SFT-teacher feedback with $\lambda \in \{0.1, 0.3, 1.0\}$.

Task 4: checkpoint success metrics

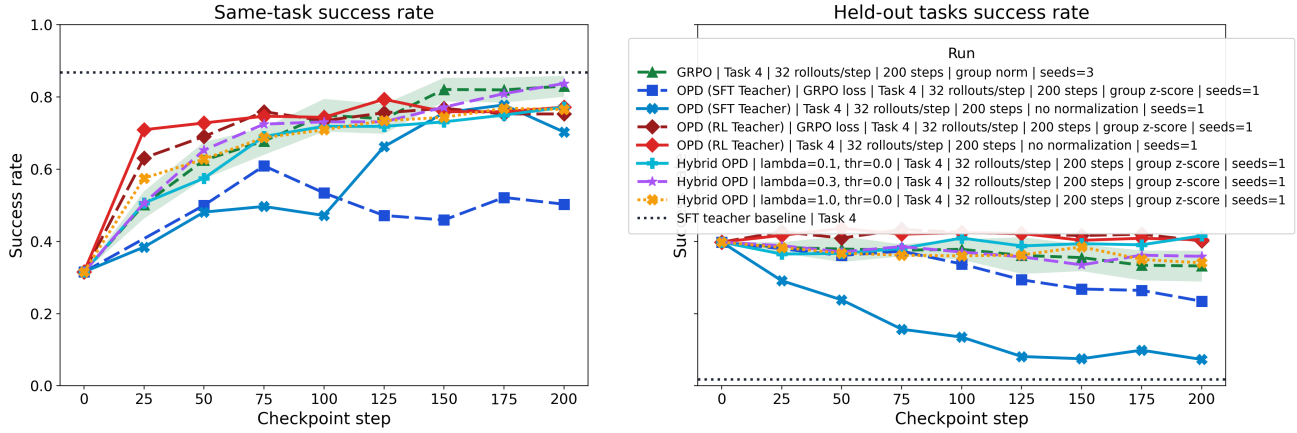


Figure 2: rps32 comparison, Task 4. Hybrid OPD uses success-gated SFT-teacher feedback with $\lambda \in \{0.1, 0.3, 1.0\}$.

Task 1: hybrid OPD rps128 checkpoint success metrics

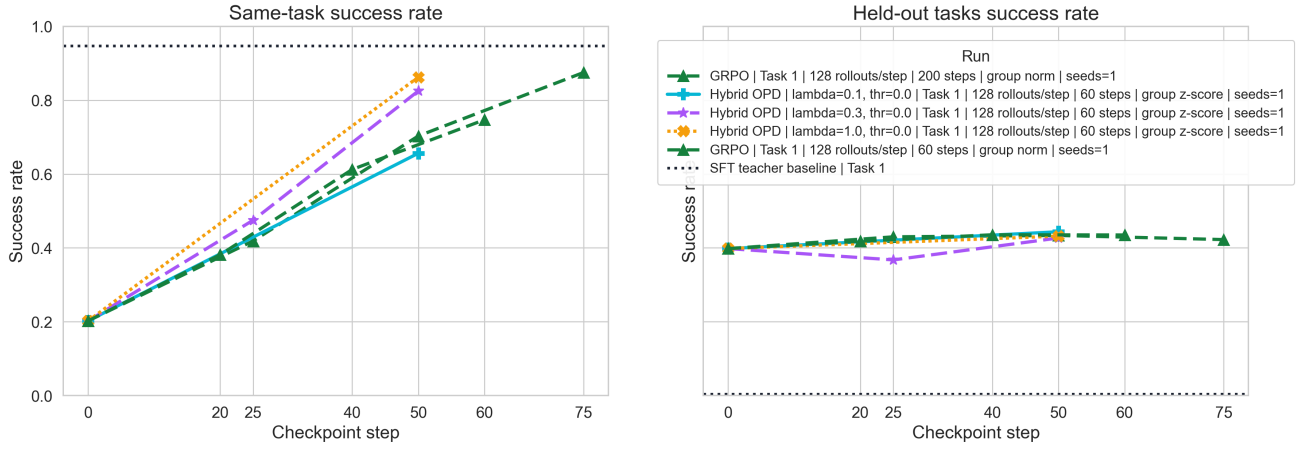


Figure 3: rps128 comparison, Task 1. Kept separate from rps32 because rollout budget differs.

Task 4: hybrid OPD rps128 checkpoint success metrics

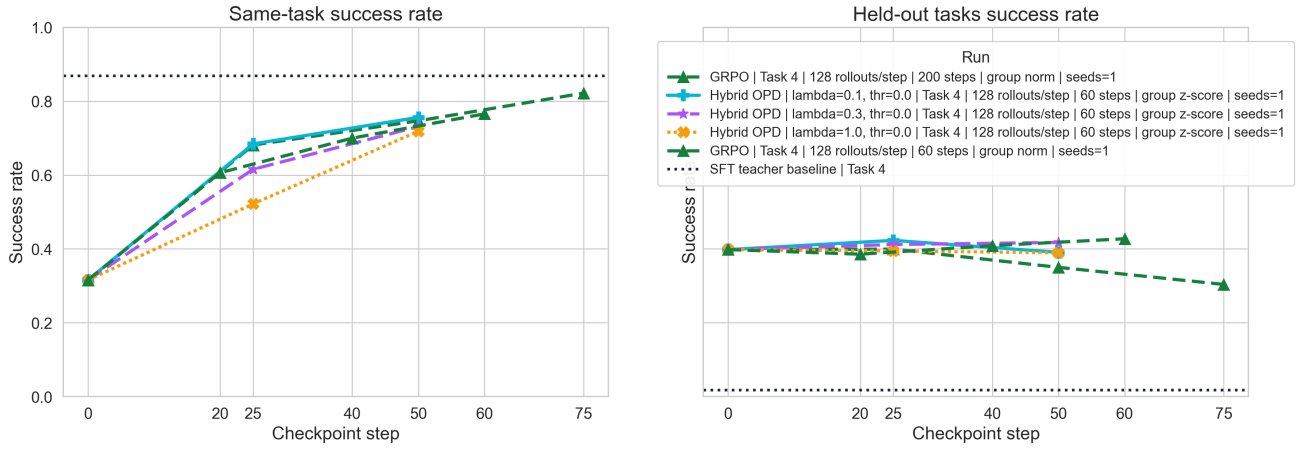


Figure 4: rps128 comparison, Task 4. Kept separate from rps32 because rollout budget differs.

Takeaways

- Hybrid OPD is implemented as *hybrid advantages* plus the GRPO clipped policy loss.
- The gate is simple: environment reward succeeds $\Rightarrow$ learn from GRPO signal; failure $\Rightarrow$ learn from teacher signal.
- Current plots support quick comparison of sample efficiency and forgetting across GRPO, OPD teacher variants, and hybrid settings.